

TASK-7

Analyze the threat of five recent DDoS attacks, and your assignment is to each select one of these incidents to investigate. You are required to research your chosen attack and prepare a brief summary covering the target, the technology used, the attacker's motive, the overall impact, and the defensive strategies that could have mitigated it.

1) Cloudflare — record 7.3 Tbps attack (mid-May 2025) :

Cloudflare reported blocking a historic DDoS that peaked at about 7.3 Tbps and delivered ~37.4 TB of data in ~45 seconds against a single IP/address, coming from over 122k source IPs. The attack was largely a UDP/volumetric flood using reflection/amplification techniques; Cloudflare's network mitigated it automatically. This event set a new bar for attack scale and reinforced the need for hyper-scale scrubbing and anycast-based mitigation.

2) Large-scale telecom industry DDoS attack (APAC, H1 2025) :

In the first half of 2025, the telecommunications sector in the Asia-Pacific region experienced one of the steepest increases in DDoS activity. According to a regional report, telecom companies were among the most heavily targeted verticals — for example, a major Chinese data-centre provider was hit with a ~2.3 Tbps attack, and a Taiwanese telecom provider suffered a sustained 11-day attack averaging ~850 Gbps.

3) Microsoft / Azure major outage attributed to DDoS (July 2024 / follow-ups in 2025 context) :

Microsoft confirmed that a major Azure / Microsoft 365 outage was initially triggered by a DDoS. The company said mitigation activities (and a staff error) then exacerbated customer impact and prolonged service disruption. That incident highlighted how even large cloud providers can be knocked into degraded states when mitigation and failover are not executed perfectly under pressure.

4) Pro-Russian hacktivist campaigns (NoName057(16) and similar) — sustained DDoS pressure (2024–2025) :

Groups like NoName057(16) ran long, frequent DDoS campaigns against Ukrainian institutions and NATO/European targets (websites, media, banks, transport). Europol/Eurojust coordinated takedowns (Operation Eastwood) in mid-2025 that disrupted their infrastructure. These campaigns show DDoS used as a geopolitical/hacktivist tool to harass, silence or distract during political events

5) Gcore Gaming-Hosting Attack (October 2025) :

In October 2025, gaming infrastructure provider Gcore was hit by a 6 Tbps DDoS attack delivering over 5 billion packets per second—one of the top-ten largest ever recorded. The short but intense UDP flood originated mainly from botnets in Brazil and the United States, briefly straining the company's defences before mitigation kicked in. The assault reflected how online gaming platforms have become prime DDoS targets, with attackers seeking to cause widespread disruption to tournaments and streaming events.

Investigation: NoName057(16) Hacktivist DDoS Campaigns (2024–2025)

Target :

The pro-Russian hacktivist group NoName057(16) primarily targeted government agencies, banks, transportation networks, and media organizations in Ukraine and NATO-aligned European countries, including Poland, Finland, Czech Republic, and Lithuania. Their campaigns often coincided with major geopolitical or military developments, such as sanctions against Russia or EU/NATO meetings supporting Ukraine. Websites of ministries, parliaments, airports, and financial institutions were frequently taken offline or degraded for hours during these coordinated DDoS waves.

Technology Used:

The group relied on Layer 7 (application-level) and volumetric DDoS attacks, using botnets and a crowd-powered platform called DDOSIA.

- DDOSIA is a volunteer-based DDoS toolkit distributed through Telegram channels. Supporters install the tool, which connects to the group's command server and launches floods against chosen targets.
- Attack methods included HTTP floods, TCP SYN floods, and DNS amplification, designed to overwhelm servers and make websites unreachable.
- The group often changed attack vectors dynamically to bypass basic firewalls and content delivery networks, showing a moderate level of technical coordination.

Attacker's Motive :

NoName057(16) is a pro-Russian hacktivist collective motivated by political ideology and propaganda goals rather than financial gain. Their operations aimed to:

1. Disrupt online communication and government services in pro-Ukraine or NATO countries.
2. Spread political narratives and show symbolic "cyber retaliation" for actions against Russia.
3. Gain visibility on social media by claiming responsibility for successful takedowns and recruiting more volunteers.
4. Their campaigns served as part of a broader information warfare strategy, reinforcing Russia's geopolitical stance and amplifying online disinformation.

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Overall Impact :

The attacks caused temporary outages and service slowdowns across multiple European government and financial websites, though critical national systems remained largely unaffected.

- Some ministries and banks experienced hours of downtime, while others were forced to block foreign traffic temporarily to mitigate impact.
- Beyond the technical disruption, the psychological and political impact was significant—creating fear, damaging public trust, and diverting cybersecurity resources from other threats.
- In mid-2025, Europol and Eurojust's Operation Eastwood dismantled parts of the DDOSIA infrastructure and arrested several operators, reducing the campaign's intensity but not fully eliminating the threat.

Defensive Strategies / Mitigation:

- Use DDoS Mitigation Services (CDN / Scrubbing Centers):
Employ providers like Cloudflare, Akamai, or Radware to absorb large-scale attacks through distributed filtering and rate limiting.
- Adaptive Layer 7 Protection:
Deploy Web Application Firewalls (WAFs) with behavioral analysis to detect HTTP flood patterns and automatically block malicious traffic.
- Geo-fencing and Rate Limiting:
During active attacks, restrict access from suspicious regions or limit request rates per IP to slow down attack flows.
- Incident Response Planning:
Maintain a DDoS response playbook that outlines escalation paths, communication protocols, and steps for rerouting traffic under pressure.